

FORMULAS

$$\text{leveraged return} = \frac{r(V_0 + V_B) - r_B V_B}{V_0}$$

rate of return for an investor after fees:

$$r = \frac{V_1 - V_0 - \text{total fees}}{V_0}$$

futures price $\approx$ spot price $\times$ (1 + risk-free rate) + storage costs – convenience yield

Standard deviation of a two-asset portfolio:

$$\sigma_{\text{portfolio}} = \sqrt{W_A^2 \sigma_A^2 + W_B^2 \sigma_B^2 + 2 W_A W_B \rho_{AB} \sigma_A \sigma_B}$$

Standard deviation of returns from historical data:

$$s^2 = \frac{\sum_{t=1}^T (R_t - \bar{R})^2}{T - 1}$$

where T = number of observations

Covariance of returns from historical data:

$$\text{Cov}_{1,2} = \frac{\sum_{t=1}^n \{ [R_{t,1} - \bar{R}_1] [R_{t,2} - \bar{R}_2] \}}{n - 1}$$

where:

$R_{t,1}$ = return on Asset 1 in period t

$R_{t,2}$ = return on Asset 2 in period t

$\bar{R}_1$ = mean return on Asset 1

$\bar{R}_2$ = mean return on Asset 2

n = number of periods

Correlation:

$$\rho_{1,2} = \frac{\text{Cov}_{1,2}}{\sigma_1 \sigma_2}$$

Beta of Asset i with the market:

$$\beta_i = \frac{[\text{Cov}(R_i, R_m)]}{\sigma_m^2} = \rho_{i,m} \left(\frac{\sigma_i}{\sigma_m} \right)$$

where $[\text{Cov}(R_i, R_m)]$ and $\rho_{i,m}$ are the covariance and correlation between the asset and the market, and σ_i and σ_m are the standard deviations of asset returns and market returns.

$$\text{equation of the CML: } E(R_P) = R_f + \left(\frac{E(R_M) - R_f}{\sigma_M} \right) \sigma_P$$

$$E(R_P) = R_f + (E(R_M) - R_f) \left(\frac{\sigma_P}{\sigma_M} \right)$$

total risk = systematic risk + unsystematic risk

Capital Asset Pricing Model (CAPM): $E(R_i) = R_f + \beta_i[E(R_{mkt}) - R_f]$

$$\text{Sharpe ratio} = \left(\frac{R_P - R_f}{\sigma_P} \right)$$

$$\text{M-squared} = R_f + \frac{\sigma_M}{\sigma_P}(R_P - R_f)$$

$$\text{Treynor measure} = \frac{R_P - R_f}{\beta_P}$$

$$\text{Jensen's alpha} = \alpha_P = R_P - [R_f + \beta_P(R_M - R_f)]$$